Transcriptomic Comparison of Plate vs Broth Growth Conditions in PHB Producing Bacteria

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Abstract

This project analyzed differential gene expression in a bacterial strain that produces a visible product on solid agar plates but not in liquid broth. The strain is suspected to belong to the actinobacterial genus *Promicromonospora*. Bacteria from this genus are known to alter metabolism and produce polyhydroxybutyrate (PHB) under nutrient or environmental stress. Using the eggNOG differential expression file (DE_results.csv) provided to me by Dr. Van Laar, I filtered significant genes based on the adjusted p values and $\log_2$ fold change thresholds, generated upregulated gene files for both conditions, and summarized the functional annotations using COG categories. A python script was developed to quantify the count of COG categories and calculate the percentages of upregulated genes for each category. Plate grown cells showed increased expression in transcriptional regulation, carbohydrate processing, and stress related pathways, while broth grown cells had higher expression of amino acid metabolism, translation, and energy production. These results suggest that plate conditions trigger stress and regulatory responses associated with the synthesis of PHB, while broth conditions support rapid growth and reduce the need for storage polymer production.

Background

Dr. Thao and her research students have isolated a bacterial strain that produces a product when grown on solid agar plates but not when grown in liquid broth. To understand why this product is expressed only on plate conditions she recruited help from Dr. Van Laar's bioinformatics students to examine differentially expressed genes between the two conditions. The strain is believed to belong to the genus *Promicromonospora* which is an actinobacterium commonly found in soil. This is a gram positive aerobic bacteria that has the ability to produce polyhydroxybutyrate (PHB). PHB is a biodegradable biopolymer that can be used as an energy reserve when the cell experiences nutrient imbalance or environmental stress.

Microbial transcriptomics analyzes the transcriptome of an organism to give insight into the metabolic pathways and regulatory systems. By comparing the differentially expressed genes on plate vs broth we can determine which genes are upregulated or downregulated to make inferences about the physiological changes the bacteria experiences in each condition.

A CSV file of the raw annotations from eggNOG tool was provided with the differential expression results. The file provided the Gene Ontology terms (GO Terms), KEGG pathways, Clusters of Orthologous Groups (COGs), and statistical measures. The functional COG categories allowed me to group the genes by biological role and identify trends across the genome such as increased transcription, changes in energy metabolism, and stress related pathways. This project analyzed the differentially expressed genes using the eggNOG annotations to determine which genes were upregulated on plates compared to broth and how these differences may contribute to PHB production.

Methods

Using my Ubuntu terminal I wrote a script to process the differential expression output file from eggNOG (DE_results.csv) and filter genes based on the adjusted p values ≤ 0.05 and log2FC ≥ 1 to identify the significantly upregulated and downregulated genes. The script generated multiple output files including:

- significant_genes.csv → genes passing the p value ≤ 0.05 cutoff
- sig_all.csv → all significant genes, not considering FC
- plate_up.csv → genes upregulated on plates (log2FC ≥ 1)
- broth up.csv → genes upregulated in broth (log2FC ≤ -1)
- top_GO.txt → GO term summary of the significant genes
- top_KEGG.txt → KEGG annotation summary of the significant genes
- top_COG.txt → COG functional category summary of the significant genes
- summary_report.txt → a text file reporting the KEGG, COG, and GO term results

These files were used for downstream functional analysis and COG categorization. After generating the filtered gene files, I wrote a Python script to analyze the COG annotations. The script imported the DE_results.csv, broth_up.csv, and plate_up.csv to count how many upregulated genes belonged to each COG category for each condition, and also to split any double COG categories. The script also calculated the percentage of total upregulated genes of each COG group and formatted the results into a table listing each COG category next to the count and percent contribution for both conditions.

Results

Differential expression analysis revealed functional differences between plate and broth grown bacteria. A total of 494 genes were upregulated on plates and 510 were upregulated in broth after applying the significance and fold change cutoffs. After grouping the COG categories, several pathways showed preference for certain conditions. Plate grown cells have the highest significant upregulation in genes involved in Function unknown (S) representing 21.90% of all plate upregulated genes. Major increases in Transcription (K) and Carbohydrate metabolism (G) followed, representing 13% and 10.50% of the plate upregulated genes.

Comparatively, broth grown cells showed strong upregulation in genes involved in Amino acid transport and metabolism (E) at 11.8%, Energy production (C) at 9.6%, and Translation (J) at 7.8% of the broth upregulated genes. Several stress and signaling related categories like O, T, and U showed higher upregulation in plates compared to the broth. This suggests that growth on solid agar plates caused a stronger environmental response. The COG distribution indicates that broth culture is favorable for growth associated pathways, while the plate culture activates transcription regulation, carbohydrate processing, and stress related functions which potentially contribute to the PHB production seen on solid medium.

Table 1. COG functional categories of genes upregulated on plate vs broth conditions

		Plate Upregulation		Broth Upregulation	
COG	**Function**	**Count**	**Percent %**	**Count**	**Percent %**
K	Transcription	64	13.00%	32	6.3%
G	Carbohydrate transport and metabolism	52	10.50%	49	9.6%
E	Amino acid transport and metabolism	44	8.90%	60	11.8%
M	Cell wall/membrane/envelope biogenesis	31	6.30%	24	4.7%
C	Energy production and conservation	31	6.30%	49	9.6%
P	Inorganic ion transport and metabolism	28	5.70%	24	4.7%
T	Signal transduction mechanisms	19	3.80%	15	2.9%
L	Replication, recombination, repair	18	3.60%	24	4.7%
I	Lipid transport and metabolism	15	3.00%	21	4.1%
O	Posttranslational modification, protein turnover, chaperones	13	2.60%	13	2.5%
J	Translation, ribosomal structure, biogenesis	13	2.60%	40	7.8%
Q	Secondary metabolite biosynthesis, transport and catabolism	13	2.60%	17	3.3%
H	Coenzyme transport and metabolism	12	2.40%	29	5.7%
F	Nucleotide transport and metabolism	9	1.80%	20	3.9%
U	Intracellular trafficking, secretion, vesicular transport	6	1.20%	4	0.8%
V	Defense mechanisms	6	1.20%	18	3.5%
D	Cell cycle control, cell division	5	1.00%	6	1.2%
N	Cell motility	1	0.20%	4	0.8%
R	General function prediction only	0	0.00%	0	0.0%
S	Function unknown	108	21.90%	89	17.5%
-	No COG assigned	38	7.70%	18	3.5%

Discussion

Members of the *Promicromonospora* genus are known to respond strongly to environmental and nutrient stresses by altering which genes their systems transcribe and by storing carbon as polymers like PHB. This bacterium diverts carbon toward storage rather than growth when resources are unbalanced. PHB production is a stress response in many actinobacteria, where excess carbon is used in storage polymers instead of being utilized in cell growth during nutrient limitation. Plate grown cells showed increased transcription, signaling pathways, and carbohydrate metabolism which are all factors that are consistent with stress adaptation. On the other hand, broth grown cells showed increased amino acid metabolism, translation, and energy production which are all characteristics of rapid growth. These differentially expressed genes suggest that PHB production occurs under plate conditions because the cells experience nutrient limitation and stress, which then triggers pathways that signal for carbon storage over active cell growth.

Conclusion

This transcriptome analysis shows that plate grown cells activate regulatory, stress response, and carbohydrate processing signaling pathways while the broth grown cells activate pathways that contribute to growth such as translation and energy metabolism. These differences explain why PHB production is observed only on solid agar plates, since the solid surface induces stress conditions forcing the bacteria to use its carbon to create storage polymers. The trends found are consistent with the known physiological behavior of other actinobacteria species of the *Promicromonospora* genus.

References

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Tatusov, R. L., Galperin, M. Y., Natale, D. A., & Koonin, E. V. (2000). The COG database: A tool for genome-scale analysis of protein functions and evolution. NCBI. https://www.ncbi.nlm.nih.gov/research/cog

Appendix

Figure 1. Complete documentation of Ubuntu code to filter differentially expressed results and create statistical output files

```
# --------Navigate to folder--------
conda env list
conda activate genome analysis
cd /mnt/c/Users/alvar/OneDrive/Desktop
cd PHB

# -------- Get significant genes with adj p val <0.05--------
awk -F, 'NR==1{print > "significant_genes.csv"; next} {if ($10+0 <= 0.05) print >
"significant_genes.csv"}' DE_results.csv

# --------Set significance and fc cutoffs--------
FILE="de_results.csv"
ALPHA=0.05       # padj cutoff
LFC=1            # |log2FC| cutoff for up/down

# --------Filter significance, plate up, broth up--------
awk -v ALPHA="$ALPHA" -v LFC="$LFC" '
BEGIN{
  OFS=","; FPAT="([^,]*)|(\"[^\"]+\")"
}
NR==1{
  for(i=1;i<=NF;i++){
    h=tolower($i)
    if(h=="padj") padj=i
    if(h=="log2foldchange") fc=i
  }
  if(!padj||!fc){ print "Missing padj or log2FoldChange header."; exit 1 }
  hdr=$0
  print hdr > "sig_all.csv"
  print hdr > "plate_up.csv"
  print hdr > "broth_up.csv"
  next
}
{
  p=$(padj)+0; f=$(fc)+0
  if(p<=ALPHA){
    print $0 >> "sig_all.csv"
    if(f>=LFC)  print $0 >> "plate_up.csv"
    if(f<=-LFC) print $0 >> "broth_up.csv"
  }
}
' "$FILE"
```

```
# --------To format my plate_up into readable columns--------
column -s, -t plate_up.csv | less -S

# --------I now have these files--------
    significant_genes.csv
    sig_all.csv
    plate_up.csv     #sig pos logfc
    broth_up.csv     #sig neg logfc

# --------Sort most common GO terms--------
cut -d',' -f15 significant_genes.csv | sort | uniq -c | sort -nr | head

# --------Sort most common KEGG pathways--------
cut -d',' -f18 significant_genes.csv | sort | uniq -c | sort -nr | head

# --------Sort most common COG categories--------
cut -d',' -f12 significant_genes.csv | sort | uniq -c | sort -nr | head

# --------Save a summary of each to a summary file--------
cut -d',' -f15 significant_genes.csv | sort | uniq -c | sort -nr > top_GO.txt
cut -d',' -f18 significant_genes.csv | sort | uniq -c | sort -nr > top_KEGG.txt
cut -d',' -f12  significant_genes.csv | sort | uniq -c | sort -nr > top_COG.txt

# --------Combine into one summary file--------
echo "==== Top GO Terms ====" > summary_report.txt
cat top_GO.txt >> summary_report.txt

echo -e "\n==== Top KEGG Pathways ====" >> summary_report.txt
cat top_KEGG.txt >> summary_report.txt

echo -e "\n==== Top COG Categories ====" >> summary_report.txt
cat top_COG.txt >> summary_report.txt

# --------I now have these additional files--------
    top_GO.txt
    top_KEGG.txt
    top_COG.txt
    summary_report.txt
```

Figure 2. Complete documentation of python script to organize the COG categories, functions, and the counts and percentages of each

```
"""
********************************************************************************
Filename: Results Table.py
Author: Isabella Fregoso
Date: 6 December 2025
Description: This module generates a functional genomics summary table showing
           which metabolic and regulatory pathways are upregulated on plate
           vs broth using COG categories.
********************************************************************************
"""
# --------Obtain the data files--------
import csv
from collections import Counter, defaultdict


# --------Load a csv & split the COG categories w double letters--------
def load_cog_counts(path, cog_column="COG_category"):
   counts = Counter()
   total = 0

   with open(path, newline='', encoding="utf-8") as f:
       reader = csv.DictReader(f)
       for row in reader:
           cog = row[cog_column].strip()
           if not cog:
               continue

           total += 1

           # Split double letter COG groups
           for letter in cog:
               if letter.isalpha() or letter == "-":
                   counts[letter] += 1

   return counts, total


# --------Import my upregulated files for broth and plate--------
plate_counts, total_plate = load_cog_counts("plate_up.csv")
broth_counts, total_broth = load_cog_counts("broth_up.csv")

# --------Master list of COG descriptions--------
cog_desc = {
   'D': "Cell cycle control, cell division",
```

```
    'M': "Cell wall/membrane/envelope biogenesis",
    'N': "Cell motility",
    'O': "Posttranslational modification, protein turnover, chaperones",
    'T': "Signal transduction mechanisms",
    'U': "Intracellular trafficking, secretion, vesicular transport",
    'V': "Defense mechanisms",
    'J': "Translation, ribosomal structure, biogenesis",
    'K': "Transcription",
    'L': "Replication, recombination, repair",
    'C': "Energy production and conservation",
    'E': "Amino acid transport and metabolism",
    'F': "Nucleotide transport and metabolism",
    'G': "Carbohydrate transport and metabolism",
    'H': "Coenzyme transport and metabolism",
    'I': "Lipid transport and metabolism",
    'P': "Inorganic ion transport and metabolism",
    'Q': "Secondary metabolite biosynthesis, transport and catabolism",
    'R': "General function prediction only",
    'S': "Function unknown",
    '-': "No COG assigned"
}

# --------Sort COG letters in group order--------
biological_order = [
    "D","M","N","O","T","U","V",
    "J","K","L",
    "C","E","F","G","H","I","P","Q",
    "R","S","-"
]

# --------Print formatted table with nice columns and headers--------
print("{:<4} {:<60} {:>8} {:>8} {:>10} {:>10}".format(
    "COG", "Function",
    "Plate#", "Plate%",
    "Broth#", "Broth%"
))
print("-" * 110)

# --------To generate the rows--------
for cog in biological_order:
    func = cog_desc.get(cog, "(unknown)")

    plate_n = plate_counts.get(cog, 0)
    broth_n = broth_counts.get(cog, 0)

# --------Calculates the percentages for each--------
    plate_pct = (plate_n / total_plate * 100) if total_plate > 0 else 0
```

```
    broth_pct = (broth_n / total_broth * 100) if total_broth > 0 else 0

    # --------Neatly prints the columns with headers--------
    print("{:<4} {:<60} {:>8} {:>7.1f}% {:>10} {:>7.1f}%".format(
        cog, func,
        plate_n, plate_pct,
        broth_n, broth_pct
    ))
# --------Prints the total of upregulated genes--------
print(f"\nTotal plate-upregulated genes: {total_plate}")
print(f"Total broth-upregulated genes: {total_broth}\n")

"""
**********************************************
              Output Table
**********************************************

COG  Function                                                        Plate#  Plate%    Broth#    Broth%
---------------------------------------------------------------------------------------------------------
D    Cell cycle control, cell division                                   5     1.0%          6      1.2%
M    Cell wall/membrane/envelope biogenesis                             31     6.3%         24      4.7%
N    Cell motility                                                       1     0.2%          4      0.8%
O    Posttranslational modification, protein turnover, chaperones       13     2.6%         13      2.5%
T    Signal transduction mechanisms                                     19     3.8%         15      2.9%
U    Intracellular trafficking, secretion, vesicular transport           6     1.2%          4      0.8%
V    Defense mechanisms                                                  6     1.2%         18      3.5%
J    Translation, ribosomal structure, biogenesis                       13     2.6%         40      7.8%
K    Transcription                                                      64    13.0%         32      6.3%
L    Replication, recombination, repair                                 18     3.6%         24      4.7%
C    Energy production and conservation                                31     6.3%         49      9.6%
E    Amino acid transport and metabolism                                44     8.9%         60     11.8%
F    Nucleotide transport and metabolism                                 9     1.8%         20      3.9%
G    Carbohydrate transport and metabolism                              52    10.5%         49      9.6%
H    Coenzyme transport and metabolism                                  12     2.4%         29      5.7%
I    Lipid transport and metabolism                                     15     3.0%         21      4.1%
P    Inorganic ion transport and metabolism                             28     5.7%         24      4.7%
Q    Secondary metabolite biosynthesis, transport and catabolism        13     2.6%         17      3.3%
R    General function prediction only                                    0     0.0%          0      0.0%
S    Function unknown                                                  108    21.9%         89     17.5%
-    No COG assigned                                                    38     7.7%         18      3.5%

Total plate-upregulated genes: 494
Total broth-upregulated genes: 510

Process finished with exit code 0
**********************************************
"""
```